Constraint Enforcement versus Residual Formulation: A Corrected Taxonomy and Selection Framework for Physics-Informed Neural Networks

Siya Patil[1], Ruchika Todi[1], Smriti Prashanth Vasisht[1], Vipul Reddy Gundalu[1], Rashmi R[1], Mallanagouda Patil[2]
[1]*B.M.S. College of Engineering*, Bengaluru, India
[2]*R V Institute of Technology and Management* , Bengaluru, India

***Abstract*—Physics-informed neural networks (PINNs) now comprise more named variants than the community can meaningfully compare. Existing classifications obscure this landscape by conflating two independent design decisions: how initial and boundary conditions are enforced—by penalty or by construction—and which residual formulation is minimised, whether strong, weak-variational, or energy. The conflation produces demonstrable misclassifications: variational methods such as the Deep Ritz method and hp-VPINNs are routinely called hard-constrained although they impose physics entirely through penalty terms. We propose a two-dimensional taxonomy separating these axes, demote network decomposition and problem setting to orthogonal descriptors, and observe that hard enforcement applies to boundary and initial conditions but essentially never to the governing equation. We classify eighteen representative methods, key six application domains to their dominant numerical difficulty, and grade the evidence behind each proposed remedy. Diagnosis proves better established than cure, and the binding constraint is comparability rather than architectural novelty: an independent review of the fluid-related PDE literature found 79% of claims of superiority over conventional solvers rested on weak baselines.**

***Index Terms*—Physics-informed neural networks, scientific machine learning, partial differential equations, taxonomy, benchmarking.**

I. Introduction

Mesh-based discretisations remain the reference tools for differential equations and, for well-posed forward problems, are difficult to beat. They are less convenient when remeshing a deforming domain dominates cost, when sparse noisy measurements must be reconciled with a model, and when coefficients must be inferred rather than prescribed.

Physics-informed neural networks (PINNs) target these cases: a network is a mesh-free ansatz for the solution, its derivatives come from automatic differentiation, and the governing equation is enforced by penalising its residual at collocation points [1]. Because the physics acts as a regulariser, training remains feasible with scarce labels, and unknown coefficients can be made trainable, placing forward and inverse problems in one framework [1], [2]. The idea predates the modern form: collocation-based neural PDE solvers appeared in 1994 [3] and a trial solution satisfying boundary conditions exactly in 1998 [4], with automatic differentiation [5] and universal approximation [6] supplying the basis. What followed 2019 was proliferation rather than consolidation: dozens of variants coexist and a practitioner choosing among them gets little guidance from the surveys that organise them (Section II), all of which inherit one conceptual problem.

The conflation. Methods are described as imposing physics *softly*, through penalty terms, or *hard*, through construction; separately, the residual minimised may be pointwise (strong), integrated against test functions (weak), or an energy functional. These are orthogonal, yet routinely merged into a single ordinal notion of "how strongly physics is enforced," and the consequence is misclassification. The Deep Ritz method [19] and hp-VPINNs [21] are grouped with hard-constraint methods because they use variational forms, yet both minimise a penalty objective and satisfy nothing by construction; conservative PINNs [22] are read as enforcing conservation exactly, yet flux continuity across interfaces enters as a weighted loss term. A taxonomy that cannot separate the Deep Ritz method from the ansatz of Lagaris *et al.* [4] is not a basis for selection.

Contributions. We separate boundary/initial-condition *enforcement* from *residual formulation* into two independent axes (Section IV, Fig. 1), resting on the observation that hard enforcement targets boundary and initial conditions and essentially never the governing equation; demote network decomposition and problem setting from axes to orthogonal descriptors; key six domains to their dominant numerical difficulty for selection guidance (Section V); and grade the evidence behind each remedy for known pathologies (Section VI). Existing reviews classify PINNs by equation class, architecture, or domain and report the same pathology list; none does either.

II. Related Surveys and Scope

Each of the seven reviews in Table I adopts a different organising principle, and none distinguishes how a constraint is imposed from the form in which it is written. Karniadakis *et al.* [2] come closest, since their inductive- and learning-bias categories align with hard and soft enforcement, but that framework classifies the *origin* of the bias, so variational

TABLE I
POSITIONING AGAINST EXISTING PINN SURVEYS. **A**: SEPARATES BC/IC ENFORCEMENT FROM RESIDUAL FORMULATION. **B**: GRADES THE EVIDENCE BEHIND PROPOSED MITIGATIONS. **C**: GIVES DOMAIN-INDEXED METHOD-SELECTION GUIDANCE.

Survey	Year	Primary organising principle	A	B	C
Karniadakis *et al.* [2]	2021	Source of the physical bias: observational, inductive, learning	Partly	No	No
Cai *et al.* [7]	2021	Single application domain (fluid mechanics)	No	No	One domain
Cuomo *et al.* [8]	2022	Methodological components and supporting theory	No	No	No
Farea *et al.* [9]	2024	Backbone architecture (ANN, CNN, RNN, GNN, attention)	No	No	No
Sophiya *et al.* [10]	2025	Equation class (ODE, PDE, fractional), then variants	No	No	No
Wong *et al.* [11]	2025	Optimisation strategy (evolutionary, gradient-free)	No	No	No
Ahmad *et al.* [12]	2026	Recent architectural variants and hyperparameter tuning	No	No	No
This work	2026	**Enforcement mechanism $\times$ residual formulation**	**Yes**	**Yes**	**Yes**

and strong-form methods fall in one cell. All seven report substantially the same pathologies without indicating which remedies have been reproduced.

Scope. This is a structured critical synthesis, not a quantitative meta-analysis: we neither re-run experiments nor aggregate reported errors, because those errors are computed on inconsistent problems and baselines, which is itself a finding (Section VI). The literature base comprises the foundational formulations; developments introducing a distinct enforcement mechanism, residual formulation, decomposition strategy, or training remedy; recent surveys, for positioning; and failure-mode, theory, and benchmark studies. Application papers evidence a domain's dominant difficulty rather than being surveyed exhaustively, and variants differing only in hyperparameters are excluded.

III. FOUNDATIONS

Consider a differential equation on Ω with boundary $\partial\Omega$ over $[0, T]$,

$$\mathcal{N}[u; \lambda](x, t) = 0, \quad x \in \Omega,\ t \in [0, T], \tag{1}$$

with $u(x, 0) = h(x)$ and $\mathcal{B}[u] = g$ on $\partial\Omega$, where $\mathcal{N}$ is a differential operator with coefficients λ and $\mathcal{B}$ may be of Dirichlet, Neumann, Robin, or periodic type. A network u_θ approximates u, and derivatives entering $\mathcal{N}$ come from reverse-mode automatic differentiation [5], so no discretisation stencil appears, though each differential order costs a further traversal of the graph. Training minimises

$$\mathcal{L}(\theta, \lambda) = w_r\mathcal{L}_r + w_b\mathcal{L}_b + w_i\mathcal{L}_i + w_d\mathcal{L}_d, \tag{2}$$

where $\mathcal{L}_r$ measures the residual of (1) at N_r collocation points, $\mathcal{L}_b$ and $\mathcal{L}_i$ the violation of boundary and initial conditions, and $\mathcal{L}_d$ misfit to observations; in the strong form $\mathcal{L}_r = N_r^{-1}\sum_i \|\mathcal{N}[u_\theta; \lambda](x_i, t_i)\|^2$. Optimisation is typically Adam followed by quasi-Newton L-BFGS [13], and ablations show training-strategy choices affect accuracy substantially [14]. Making λ trainable turns (2) into an inverse problem without an adjoint [1].

IV. A CORRECTED TAXONOMY

Fig. 1 summarises the proposed two-dimensional taxonomy, the representative occupancy of its cells, and the two descriptors we argue are not axes.

‘‘‘

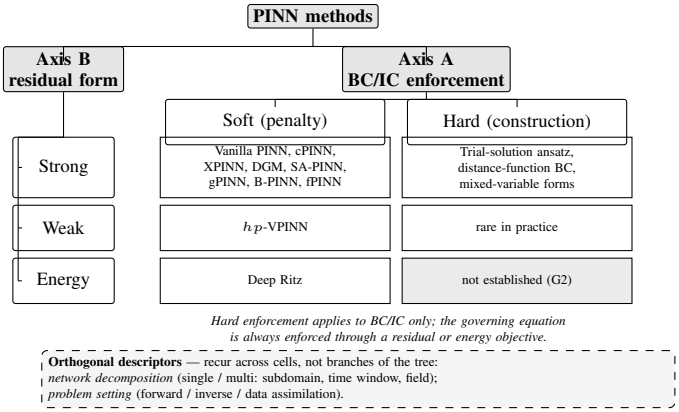


Fig. 1. The proposed taxonomy. Axes A and B are independent, so methods occupy cells of their product rather than branches of one hierarchy. Cell entries are representative; the shaded cell is unoccupied (gap G2).

A. Axis A: Enforcement of Boundary and Initial Conditions

Under *soft* enforcement, $\mathcal{L}_b$ and $\mathcal{L}_i$ are penalty terms. This applies to any operator and boundary type, which explains its dominance, but w_b, w_i must be chosen, and since gradient magnitudes of the terms can differ substantially a poor choice lets one term dominate the update direction [15]. Nothing forces $\mathcal{L}_b$ to zero, so a converged model can satisfy the interior equation while violating its boundary data.

Under *hard* enforcement the constraint holds by construction. The original device is $\hat{u}(x) = g(x) + d(x)\mathcal{N}(x; \theta)$ with g interpolating the boundary data and d vanishing on $\partial\Omega$ [4]; the obstacle is constructing d for non-trivial geometry, which approximate distance fields from R-functions address [16]. Related constructions use auxiliary networks in a mixed-variable formulation [17] or constrained formulations for inverse design [18]. Hard enforcement removes w_b from tuning and guarantees boundary consistency, at the cost of generality, since each geometry and boundary type needs its own construction.

Crucially, *hard enforcement applies to boundary and initial conditions, not to the governing equation*: imposing $\mathcal{N}[u_\theta] \equiv 0$ for all θ would mean constructing an exact solution. Every method here therefore enforces (1) softly, and claims that a method "enforces the physics exactly" should be read as claims about boundary, initial, or interface conditions. This

resolves most of the misclassifications above.

B. Axis B: Residual Formulation

The *strong* form evaluates the operator pointwise, requiring differentiability to the order of $\mathcal{N}$ and remaining sensitive to solution roughness. The *weak* form multiplies the residual by test functions and integrates by parts, lowering that requirement and shifting cost from differentiation to quadrature; hp-VPINNs use localised polynomial test functions [21]. The *energy* form replaces the residual with a functional whose stationary point is the solution, as in the Deep Ritz method [19], and exists only where a variational principle does, which excludes many transport problems. The Deep Galerkin method [20], despite its name, minimises a strong-form residual. Axis B changes conditioning, smoothness requirements, and quadrature cost, not whether a constraint holds by construction.

C. Orthogonal Descriptors, Not Axes

Network decomposition into subdomains, time windows, or fields is a scalability and conditioning strategy, not a statement about enforcement. Spatial decomposition [22], [23], overlapping partitions with local bases [24], and temporal windowing [27] retain the enforcement and residual choices of their single-network counterparts, and their principal payoff is parallelism [25]. It introduces interface conditions that are themselves enforced softly, so it interacts with Axis A without being part of it; nor is it free, since an analysis of XPINNs finds it raises expressivity but lowers the data per subdomain, so generalisation improves only under conditions on the solution and the partition [26].

Problem setting—forward solution, coefficient inference, data assimilation—is a property of the problem: the same vanilla PINN addresses all three depending on whether λ is trainable and $\mathcal{L}_d$ present. Treating it as an axis yields cells describing problems rather than methods and forces orthogonal capabilities such as Bayesian inference over θ [31] into arbitrary positions, so Table II records it as an attribute. *Operator learning* likewise sits outside both axes in its base form: DeepONet [33] and the Fourier neural operator [34] learn solution operators from paired data and impose no physical constraint, whereas the physics-informed DeepONet adds a residual penalty and so enters the soft–strong cell [35]. Calling an unconstrained operator learner physics-constrained inverts its status.

V. Application Domains and Method Selection

Surveys usually organise applications by listing the variants tried in each field, which is unhelpful for selection because the variant used often reflects the authors' familiarity rather than the problem's requirements. We instead key each domain to its dominant numerical difficulty, a property of the equations, and read the indicated features off Section IV (Table III).

Three patterns follow. First, the dominant difficulty predicts the useful axis: where the obstacle is the boundary data itself, as in solid mechanics with prescribed tractions, Axis A matters and hard construction pays [16], [17]; where it is oscillatory or rough structure, as in high-wavenumber electromagnetics and seismic problems, Axis B and the input embedding matter while enforcement is largely irrelevant; where it is domain size or time horizon, neither axis helps and decomposition is the lever [23]–[25]. Second, reported successes concentrate in inverse and assimilation settings: velocity and pressure from sparse visualisation [45], near-wall haemodynamics [48], cardiac activation maps with uncertainty [49], conductivity from interior temperatures [46], and material parameters in solids [47] all involve problems where a conventional solver needs an adjoint or repeated forward solves; PINNs are competitive where the classical route is awkward, not merely where it is expensive. Third, forward-solution claims are correspondingly the weakest part of this literature (Section VI). The default is therefore to prefer hard enforcement where boundary data must hold exactly and the geometry admits a distance construction, to add decomposition only once a single network has saturated, and to treat forward-only use as requiring justification against a conventional solver.

VI. Failure Modes and Strength of Evidence

A remedy validated only by its proposers on their own benchmark carries different weight from one reproduced in a third-party comparison, so Table IV applies a four-level grading; the pattern it reveals is that diagnosis is on firmer ground than remedy. That loss-term gradients are imbalanced [15], and that this is visible in the associated kernel spectra [37], is well established, and a third-party benchmark confirms the weighting scheme materially affects accuracy [43]. Which scheme to prefer is less settled, and the most extensive ablations [14] come from the group that proposed several of them—which is why we grade diagnosis more confidently than any individual remedy. Spectral bias is likewise established independently of PINNs [38] and Fourier embeddings help [39], but the bandwidth remains problem-dependent. Adaptive collocation is best supported, because [40] is a systematic comparison across strategies rather than an advocacy study.

The most consequential entry is the last. McGreivy and Hakim [44] reviewed machine learning for fluid-related PDEs and found that 79% of articles claiming to outperform a standard numerical method—sixty of seventy-six—compared against a weak baseline, with further evidence of outcome-reporting and publication bias. This concerns the evidential basis of the forward-solution literature rather than PINNs' potential, and it means aggregating reported speedups would propagate the bias rather than measure it, which is why this paper grades evidence instead of averaging numbers. Independently, PINNacle [43] evaluates roughly ten methods on more than twenty PDEs spanning complex geometry, multi-scale behaviour, nonlinearity, and high dimensionality, supplying the shared basis for comparison that individual studies do not. The defensible position is therefore narrower but firmer: PINNs are established where the classical alternative is structurally awkward—coefficient inference without an adjoint, assimilation of sparse noisy data, mesh-free treatment of

TABLE II

EIGHTEEN REPRESENTATIVE METHODS UNDER THE CORRECTED TAXONOMY. ENFORCEMENT: S SOFT, H HARD, N NONE. FORM: S STRONG, W WEAK, E ENERGY. NETS: 1 SINGLE, M MULTIPLE. SETTING: F FORWARD, I INVERSE, U UNCERTAINTY. LIMITATIONS ARE THOSE REPORTED BY THE CITED WORK OR BY THE STUDIES OF SECTION VI.

Method	Enf.	Form	Nets	Set.	Dominant reported limitation	Ref.
Vanilla PINN	S	S	1	F,I	Loss balancing; spectral bias; convection-dominated	[1]
Trial-solution ansatz	H	S	1	F	Constructing the vanishing function for general geometry	[4]
Exact-BC constructions	H	S	1/M	F,I	Geometry-specific; distance field or aux. nets cost	[16]–[18]
Deep Galerkin method	S	S	1	F	Shares strong-form penalty pathologies	[20]
Deep Ritz method	S	E	1	F	Requires an existing variational principle	[19]
hp-VPINN	S	W	1	F	Test-function and quadrature design to tune	[21]
cPINN	S	S	M	F	Conservation approximate; interface weights to tune	[22], [25]
XPINN	S	S	M	F,I	Interface tuning; gain conditional on the partition	[23], [26]
FBPINN	S	S	M	F	Partition and window design; overlap cost	[24]
Temporal windowing	S	S	M	F	Error accumulates across successive windows	[27]
Causal weighting	S	S	1	F	Adds a weighting schedule to tune	[28]
SA-PINN	S	S	1	F,I	Adds per-point optimisation variables	[29]
gPINN	S	S	1	F,I	Higher differentiation cost per step	[30]
B-PINN	S	S	1	F,I,U	Cost of posterior sampling over point estimation	[31]
fPINN	S	S	1	F,I	Auxiliary points grow with dimension	[32]
Operator learning	N	—	M	F	Needs paired data; no physical guarantee	[33], [34]
Physics-informed DeepONet	S	S	M	F	Inherits PINN training pathologies	[35]
PIKAN	S	S	1	F,I	Sensitive to seed and basis order	[36]

TABLE III

DOMAIN-INDEXED SELECTION MAP. DOMAINS ARE KEYED TO THEIR DOMINANT NUMERICAL DIFFICULTY RATHER THAN TO THE VARIANTS THAT HAPPEN TO HAVE BEEN TRIED. INDICATED FEATURES ARE INFERRED FROM SECTION IV; THEY ARE DESIGN GUIDANCE, NOT CLAIMS ABOUT PUBLISHED RESULTS.

Domain	Equations	Dominant difficulty	Indicated features (Sec. IV)	Studies
Fluid mechanics	Navier–Stokes, Euler	Convection dominance; multi-scale structure	Strong form with causal ordering; decomposition for large domains	[7], [45]
Heat transfer	Heat, Poisson, energy	Coefficient inference from interior data	Soft strong form, trainable λ; energy form if elliptic	[46]
Solid mechanics	Elastostatics, elastodynamics	Exact traction and displacement data	Hard BC construction; mixed-variable; energy form	[17], [47]
Biomedical	Navier–Stokes, Eikonal	Sparse noisy measurement; uncertainty	Assimilation via $\mathcal{L}_d$; Bayesian treatment	[48], [49]
Geophysics	Acoustic, elastic wave	Large domains, long horizons; high frequency	Spatial decomposition; temporal windowing; embeddings	[50]
Electromagnetics	Maxwell, Helmholtz	High wavenumber; oscillatory solutions	Frequency-aware embeddings; weak form	[51]

TABLE IV

FAILURE MODES WITH GRADED EVIDENCE FOR THEIR REMEDIES. **E1**: ASSESSED IN AN INDEPENDENT MULTI-METHOD BENCHMARK OR SYSTEMATIC COMPARISON. **E2**: CORROBORATED BY GROUPS OTHER THAN THE ORIGINATING ONE. **E3**: EVIDENCE CHIEFLY FROM THE ORIGINATING STUDY. **T**: THEORETICAL RESULT WITHOUT BROAD EMPIRICAL VALIDATION.

Failure mode	Mechanism	Representative mitigations	Evidence
Loss-term imbalance	Loss-term gradients differ by orders of magnitude	Annealing [15]; kernel weighting [37]; self-adaptive weights [29]	E1: benchmark [43]; ablations [14]
Spectral bias	Low frequencies fitted first; slow oscillatory convergence	Fourier feature embeddings [39]; decomposition [24]	E2: established independently [38]
Causality violation	A global residual lets late times fit before early ones	Temporal windowing [27]; causal weighting [28]	E2 windowing; E3 weighting
Collocation placement	Uniform sampling under-resolves sharp gradients	Residual-based adaptive refinement [13] and variants [40]	E1: systematic comparison [40]
Convection-dominated failure	Loss landscape ill-conditions as transport grows	Curriculum and sequence-to-sequence training [41]	E1: reproduced in [43]
No accuracy guarantee	Bounds need attained error and collocation density	Generalisation-error estimates [42]	T
Weak-baseline evaluation	Baselines are under-tuned conventional solvers	Standardised benchmarks [43]; reporting protocols [44]	E1

complex geometry—whereas for well-posed forward problems on tractable meshes the evidence does not support a claim of preference.

VII. Research Gaps

The taxonomy and the evidence grading together identify five gaps more specific than the customary appeals for better loss balancing and uncertainty quantification. *G1: the hard-constraint branch is under-evaluated.* Table II holds two hard-enforcement entries against fifteen soft, and no hard method appears among the graded remedies; since hard enforcement removes the boundary weights from tuning entirely, the decisive comparison—hard against carefully tuned soft enforcement, on identical problems and budgets—is largely absent. *G2: the hard–energy cell is unoccupied.* Fig. 1 shows no established method combining exact boundary construction with an energy formulation, although elastostatics offers both a variational principle [19] and a need for exact boundary data [17]; whether this is obstructed or merely unexplored is open. *G3: remedies are evaluated singly.* Adaptive weighting, frequency embeddings, adaptive sampling, and causal ordering are each assessed against a vanilla baseline, yet the ablations of Wang *et al.* [14] indicate the choices are not independent, so gains cannot be assumed additive. *G4: reporting practice is the binding constraint.* Given the prevalence of weak baselines [44], the highest-value contribution is not another variant but a reporting protocol—baseline tuning effort, hardware, wall-clock cost, seed variance, and error definition stated together—with evaluation on shared benchmarks [43]. *G5: comparability across the constrained/unconstrained boundary.* Operator learners [33], [34] amortise cost across a parameter family while PINNs solve one instance, so per-solve comparisons are not like-for-like; physics-informed operator learning [35] and KAN-based representations [36] blur that boundary, and the latter's sensitivity to seed and basis order suggests seed variance be reported as standard.

VIII. Conclusion

Separating boundary-condition enforcement from residual formulation removes a class of category errors: variational methods described as hard-constrained, conservative PINNs as exactly conservative, and unconstrained operator learners as physics-constrained. Hard enforcement applies to boundary and initial conditions and essentially never to the governing equation, and decomposition and problem setting are descriptors rather than axes. Applying the taxonomy to eighteen methods and six domains yields selection guidance keyed to a problem's dominant difficulty rather than to precedent, and grading the evidence shows diagnosis is better established than cure. With 79% of claimed outperformance in the fluid-related PDE literature resting on weak baselines [44], further variants are of limited value until comparison practice improves.

References

[1] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear PDEs," *J. Comput. Phys.*, vol. 378, pp. 686–707, 2019.

[2] G. E. Karniadakis *et al.*, "Physics-informed machine learning," *Nat. Rev. Phys.*, vol. 3, no. 6, pp. 422–440, 2021.

[3] M. W. M. G. Dissanayake and N. Phan-Thien, "Neural-network-based approximations for solving partial differential equations," *Commun. Numer. Methods Eng.*, vol. 10, no. 3, pp. 195–201, 1994.

[4] I. E. Lagaris, A. Likas, and D. I. Fotiadis, "Artificial neural networks for solving ordinary and partial differential equations," *IEEE Trans. Neural Netw.*, vol. 9, no. 5, pp. 987–1000, 1998.

[5] A. G. Baydin *et al.*, "Automatic differentiation in machine learning: A survey," *J. Mach. Learn. Res.*, vol. 18, no. 153, pp. 1–43, 2018.

[6] K. Hornik, M. Stinchcombe, and H. White, "Multilayer feedforward networks are universal approximators," *Neural Netw.*, vol. 2, no. 5, pp. 359–366, 1989.

[7] S. Cai *et al.*, "Physics-informed neural networks (PINNs) for fluid mechanics: A review," *Acta Mech. Sin.*, vol. 37, no. 12, pp. 1727–1738, 2021.

[8] S. Cuomo *et al.*, "Scientific machine learning through physics-informed neural networks: Where we are and what's next," *J. Sci. Comput.*, vol. 92, no. 3, p. 88, 2022.

[9] A. Farea, O. Yli-Harja, and F. Emmert-Streib, "Understanding physics-informed neural networks: Techniques, applications, trends, and challenges," *AI*, vol. 5, no. 3, pp. 1534–1557, 2024.

[10] A. Ajithkumar Sophiya *et al.*, "A comprehensive analysis of PINNs: Variants, applications, and challenges," *arXiv:2505.22761*, 2025.

[11] J. C. Wong *et al.*, "Evolutionary optimization of physics-informed neural networks: Evo-PINN frontiers and opportunities," *arXiv:2501.06572*, 2025.

[12] Ahmad *et al.*, "Evolution of physics-informed neural networks: Recent architectural variants and optimization strategies," *Array*, vol. 29, p. 100688, 2026.

[13] L. Lu *et al.*, "DeepXDE: A deep learning library for solving differential equations," *SIAM Rev.*, vol. 63, no. 1, pp. 208–228, 2021.

[14] S. Wang *et al.*, "An expert's guide to training physics-informed neural networks," *arXiv:2308.08468*, 2023.

[15] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient flow pathologies in physics-informed neural networks," *SIAM J. Sci. Comput.*, vol. 43, no. 5, pp. A3055–A3081, 2021.

[16] N. Sukumar and A. Srivastava, "Exact imposition of boundary conditions with distance functions in physics-informed deep neural networks," *Comput. Methods Appl. Mech. Eng.*, vol. 389, 2022.

[17] C. Rao, H. Sun, and Y. Liu, "Physics-informed deep learning for computational elastodynamics without labeled data," *J. Eng. Mech.*, vol. 147, no. 8, p. 04021043, 2021.

[18] L. Lu *et al.*, "Physics-informed neural networks with hard constraints for inverse design," *SIAM J. Sci. Comput.*, vol. 43, no. 6, pp. B1105–B1132, 2021.

[19] W. E and B. Yu, "The deep Ritz method: A deep learning-based numerical algorithm for solving variational problems," *Commun. Math. Stat.*, vol. 6, no. 1, pp. 1–12, 2018.

[20] J. Sirignano and K. Spiliopoulos, "DGM: A deep learning algorithm for solving partial differential equations," *J. Comput. Phys.*, vol. 375, pp. 1339–1364, 2018.

[21] E. Kharazmi, Z. Zhang, and G. E. Karniadakis, "hp-VPINNs: Variational physics-informed neural networks with domain decomposition," *Comput. Methods Appl. Mech. Eng.*, vol. 374, p. 113547, 2021.

[22] A. D. Jagtap, E. Kharazmi, and G. E. Karniadakis, "Conservative physics-informed neural networks on discrete domains for conservation laws," *Comput. Methods Appl. Mech. Eng.*, vol. 365, p. 113028, 2020.

[23] A. D. Jagtap and G. E. Karniadakis, "Extended physics-informed neural networks (XPINNs): A generalized space-time domain decomposition based deep learning framework for nonlinear PDEs," *Commun. Comput. Phys.*, vol. 28, no. 5, pp. 2002–2041, 2020.

[24] B. Moseley, A. Markham, and T. Nissen-Meyer, "Finite basis physics-informed neural networks (FBPINNs): A scalable domain decomposition approach," *Adv. Comput. Math.*, vol. 49, no. 4, p. 62, 2023.

[25] K. Shukla, A. D. Jagtap, and G. E. Karniadakis, "Parallel physics-informed neural networks via domain decomposition," *J. Comput. Phys.*, vol. 447, p. 110683, 2021.

[26] Z. Hu *et al.*, "When do extended physics-informed neural networks (XPINNs) improve generalization?," *SIAM J. Sci. Comput.*, vol. 44, no. 5, pp. A3158–A3182, 2022.

[27] C. L. Wight and J. Zhao, "Solving Allen–Cahn and Cahn–Hilliard equations using the adaptive physics-informed neural networks," *Commun. Comput. Phys.*, vol. 29, no. 3, pp. 930–954, 2021.

[28] S. Wang, S. Sankaran, and P. Perdikaris, "Respecting causality for training physics-informed neural networks," *Comput. Methods Appl. Mech. Eng.*, vol. 421, p. 116813, 2024.

[29] L. D. McClenny and U. M. Braga-Neto, "Self-adaptive physics-informed neural networks," *J. Comput. Phys.*, vol. 474, p. 111722, 2023.

[30] J. Yu *et al.*, "Gradient-enhanced physics-informed neural networks for forward and inverse PDE problems," *Comput. Methods Appl. Mech. Eng.*, vol. 393, p. 114823, 2022.

[31] L. Yang, X. Meng, and G. E. Karniadakis, "B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data," *J. Comput. Phys.*, vol. 415, p. 109913, 2021.

[32] G. Pang, L. Lu, and G. E. Karniadakis, "fPINNs: Fractional physics-informed neural networks," *SIAM J. Sci. Comput.*, vol. 41, no. 4, pp. A2603–A2626, 2019.

[33] L. Lu *et al.*, "Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators," *Nat. Mach. Intell.*, vol. 3, no. 3, pp. 218–229, 2021.

[34] Z. Li *et al.*, "Fourier neural operator for parametric partial differential equations," in *Proc. ICLR*, 2021.

[35] S. Wang, H. Wang, and P. Perdikaris, "Learning the solution operator of parametric PDEs with physics-informed DeepONets," *Sci. Adv.*, vol. 7, no. 40, p. eabi8605, 2021.

[36] K. Shukla *et al.*, "A comprehensive and FAIR comparison between MLP and KAN representations for differential equations and operator networks," *Comput. Methods Appl. Mech. Eng.*, vol. 431, p. 117290, 2024.

[37] S. Wang, X. Yu, and P. Perdikaris, "When and why PINNs fail to train: A neural tangent kernel perspective," *J. Comput. Phys.*, vol. 449, p. 110768, 2022.

[38] N. Rahaman *et al.*, "On the spectral bias of neural networks," in *Proc. ICML*, PMLR 97, pp. 5301–5310, 2019.

[39] M. Tancik *et al.*, "Fourier features let networks learn high frequency functions in low dimensional domains," in *Proc. NeurIPS*, vol. 33, pp. 7537–7547, 2020.

[40] C. Wu *et al.*, "A comprehensive study of non-adaptive and residual-based adaptive sampling for physics-informed neural networks," *Comput. Methods Appl. Mech. Eng.*, vol. 403, p. 115671, 2023.

[41] A. S. Krishnapriyan *et al.*, "Characterizing possible failure modes in physics-informed neural networks," in *Proc. NeurIPS*, vol. 34, pp. 26548–26560, 2021.

[42] S. Mishra and R. Molinaro, "Estimates on the generalization error of physics-informed neural networks for approximating a class of inverse problems for PDEs," *IMA J. Numer. Anal.*, vol. 42, no. 2, pp. 981–1022, 2022.

[43] Z. Hao *et al.*, "PINNacle: A comprehensive benchmark of physics-informed neural networks for solving PDEs," in *Proc. NeurIPS Datasets and Benchmarks Track*, 2024.

[44] N. McGreivy and A. Hakim, "Weak baselines and reporting biases lead to overoptimism in machine learning for fluid-related partial differential equations," *Nat. Mach. Intell.*, vol. 6, no. 10, pp. 1256–1269, 2024.

[45] M. Raissi, A. Yazdani, and G. E. Karniadakis, "Hidden fluid mechanics: Learning velocity and pressure fields from flow visualizations," *Science*, vol. 367, no. 6481, pp. 1026–1030, 2020.

[46] S. Cai *et al.*, "Physics-informed neural networks for heat transfer problems," *J. Heat Transfer*, vol. 143, no. 6, p. 060801, 2021.

[47] E. Haghighat *et al.*, "A physics-informed deep learning framework for inversion and surrogate modeling in solid mechanics," *Comput. Methods Appl. Mech. Eng.*, vol. 379, p. 113741, 2021.

[48] A. Arzani, J.-X. Wang, and R. M. D'Souza, "Uncovering near-wall blood flow from sparse data with physics-informed neural networks," *Phys. Fluids*, vol. 33, no. 7, p. 071905, 2021.

[49] F. Sahli Costabal *et al.*, "Physics-informed neural networks for cardiac activation mapping," *Front. Phys.*, vol. 8, p. 42, 2020.

[50] C. Song and T. Alkhalifah, "Solving the frequency-domain acoustic VTI wave equation using physics-informed neural networks," *Geophys. J. Int.*, vol. 225, no. 2, pp. 846–859, 2021.

[51] Y. Chen *et al.*, "Physics-informed neural networks for inverse problems in nano-optics and metamaterials," *Opt. Express*, vol. 28, no. 8, pp. 11618–11633, 2020.